

MMET 410 – Lab 3

Fall 2026

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**Set up RSLogix 5000 software modules

**Have the worksheet pulled out on the side!

Lab 3A

1. Draw 2 diagrams and download them to machine.
1. Answer the following questions Q1 and Q2 for each task.
1. Only need to include the question and answer in your report. (Don't need the sequence of actions, tag table, and diagram)

Lab report format - 3B

- a. Problem Statements – Task Number
- b. Sequence of operations . Cover all possible branches
 1. LS2 is closed/on. Push Start button, then
 2. LS1 is closed/on. then
 3. LS2 is closed/on. then ...
 4. ...
 5. Push stop button, then

Lab report format - 3B

- c. Input and output devices and terminal address assignments

Tag name and corresponding address

For example:

Start	Local.8.I.Data.0
Stop	
LS1	

- d. Ladder logic program
 - Screenshot
 - Focus on main part.

Lab 3B-Task 1

1. Start bit =1 -> System ON
 **Stop bit = 0
 1. System ON -> LS2 bit = 1 -> Conveyor FORWARD (CR1 ON)
 **LS1 bit = 0
 1. System ON -> LS1 bit = 1 -> Conveyor BACKWARD (CR2 ON)
 **LS2 bit = 0
 1. Stop bit = 1 -> System OFF
- 3 rungs: **rung 0 is for system / rung 1 is for CR1 / rung 2 is for CR2**
 - 4 inputs: Start / Stop / LS1 / LS2
 - 3 outputs: System / CR1 / CR2

Lab 3B – Task 2: Test

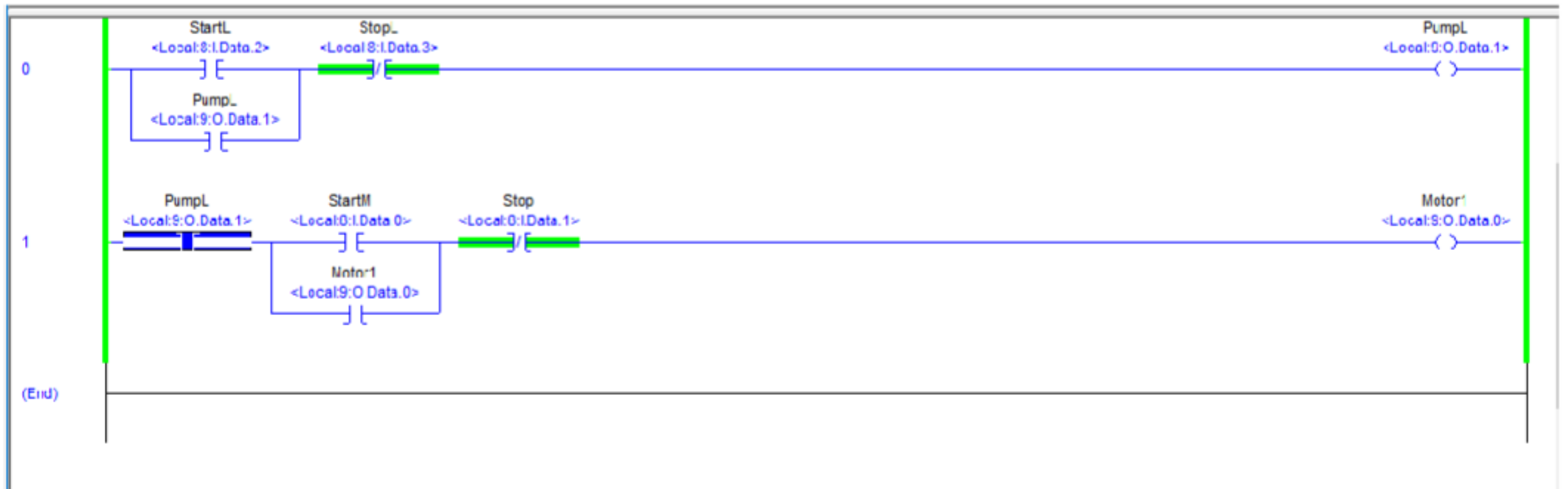
Start state: DOWN_LIMIT = On | UP_LIMIT = Off | no button pressed

Check before power-up: 5 rungs · exactly three examine-offs per motion rung · interlock uses the opposite motor OUTPUT, not the button · OTE, not OTL/OTU · no duplicate output coils

#	Action	Expected result	If it fails, look here
T1	Touch nothing	CLOSED lit; AJAR and OPEN off; all four motor outputs off	AJAR also lit — rung 0 is missing the DOWN_LIMIT examine-off
T2	Tap DOOR_UP and release	MOTOR_UP + MOTOR_FORWARD stay on after you let go	Drops on release — no seal-in branch in rung 3
T3	Flip DOWN_LIMIT off	CLOSED off, AJAR on, motor still running	AJAR stays off — rung 0 logic is reversed
T4	Flip UP_LIMIT on	Motor stops at once; OPEN on, AJAR off	Motor keeps running — the up rung has no UP_LIMIT examine-off
T5	With UP_LIMIT still on, tap DOOR_UP	Nothing happens	It restarts — the limit is only in the seal-in branch, not in series
T6	UP_LIMIT off, tap DOOR_DOWN, then flip DOWN_LIMIT on	Seals in, runs down, stops at the bottom; CLOSED on	The down rung is not a true mirror of the up rung
T7	Restart upward, then press STOP mid-travel	All motor outputs drop; AJAR stays on; nothing restarts on release	Keeps running — STOP sits inside the parallel branch, not in series
T8	Restart upward, press DOOR_DOWN while it is moving	Nothing changes — still going up, MOTOR_DN and DOWNWARD stay off	Both directions on = no interlock; direction flips = interlock misplaced
T9	Press STOP, then press DOOR_DOWN	Now it runs down normally	Still dead — the interlock uses the button instead of the motor output

Lab 3B-Task 1

- Spring return !!!
- Conditional activation !!!
- Same as what you did in task 4 lab 2



Lab 3B-Task 1

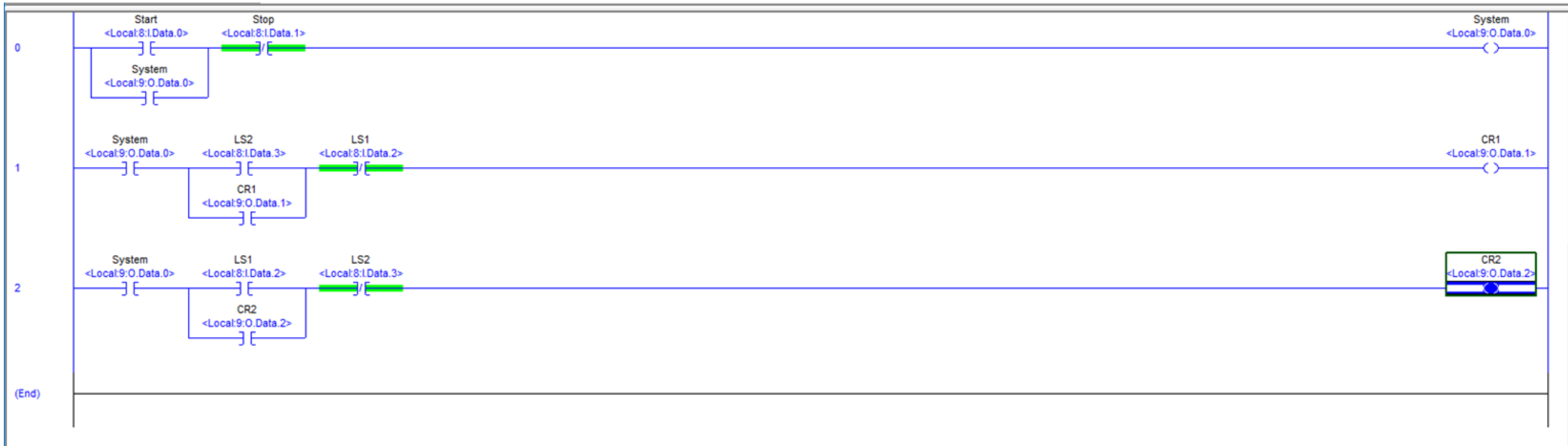
Tags and addresses

IN-0 means local.8.I.Data.0 **start** is tag name

OUT-0 means local.9.O.Data.0 **system** is tag name

- For input, use **IN-0** as **start**, **IN-1** as **stop**, **IN-2** as **LS1**, **IN-3** as **LS2**
- For output, use **OUT-0** as **system**, **OUT-1** as **CR1**, **OUT-2** as **CR2**

Lab 3B-Task 1



Lab 3B – Task 2: Input Devices

3 pushbuttons + 2 toggle switches

Address	Tag	Hardware type	Notes
IN-0	DOOR_UP	Pushbutton (momentary, NO, spring return)	Bit is 1 only while pressed — held on by the MOTOR_UP seal-in
IN-1	DOOR_DOWN	Pushbutton	Same as above — held on by the MOTOR_DN seal-in
IN-2	STOP_DOOR	Pushbutton	Programmed as XIO — the rung is true only when it is NOT pressed
IN-4	UP_LIMIT	Toggle switch (maintained)	Flip on by hand to simulate the door reaching the top
IN-5	DOWN_LIMIT	Toggle switch (maintained)	Flip on by hand to simulate the door reaching the bottom

Why IN-3 is skipped: IN-3 is still a pushbutton, but a limit sensor has to hold its state — the DOOR_OPEN light must stay lit while the door sits at the top. So the two sensors move up to IN-4 and IN-5, the first maintained toggle switches on the trainer.

Lab 3B – Task 2: Test

Start state: DOWN_LIMIT = On | UP_LIMIT = Off | no button pressed

Check before power-up: 5 rungs · exactly three examine-offs per motion rung · interlock uses the opposite motor OUTPUT, not the button · OTE, not OTL/OTU · no duplicate output coils

#	Action	Expected result	If it fails, look here
T1	Touch nothing	CLOSED lit; AJAR and OPEN off; all four motor outputs off	AJAR also lit — rung 0 is missing the DOWN_LIMIT examine-off
T2	Tap DOOR_UP and release	MOTOR_UP + MOTOR_FORWARD stay on after you let go	Drops on release — no seal-in branch in rung 3
T3	Flip DOWN_LIMIT off	CLOSED off, AJAR on, motor still running	AJAR stays off — rung 0 logic is reversed
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T5	With UP_LIMIT still on, tap DOOR_UP	Nothing happens	It restarts — the limit is only in the seal-in branch, not in series
T6	UP_LIMIT off, tap DOOR_DOWN, then flip DOWN_LIMIT on	Seals in, runs down, stops at the bottom; CLOSED on	The down rung is not a true mirror of the up rung
T7	Restart upward, then press STOP mid-travel	All motor outputs drop; AJAR stays on; nothing restarts on release	Keeps running — STOP sits inside the parallel branch, not in series
T8	Restart upward, press DOOR_DOWN while it is moving	Nothing changes — still going up, MOTOR_DN and DOWNWARD stay off	Both directions on = no interlock; direction flips = interlock misplaced
T9	Press STOP, then press DOOR_DOWN	Now it runs down normally	Still dead — the interlock uses the button instead of the motor output

Lab 3B-Task 2

- 5 inputs: 3 Buttons: use IN-0 as Door_UP, IN-1 as Door_DN, IN-2 as Stop
2 Switches to represent sensors: IN-4 as UP_limit, IN-5 as DN_limit
- 7 outputs:
OUT-0 as AJAR, OUT-1 as Door_OPEN, OUT-2 as Door_CLOSED
OUT-3 as Motor_UP, OUT-4 as Motor_DN,
OUT-5 as Motor_UPWARD, OUT-6 as Motor DOWNWARD
- **5 rungs, first status lights three rungs are for(AJAR/OPEN/CLOSED), the other two rungs are for Motor UPWARD/DOWNWARD actions**
- HINT: 3 examine off instructions for each of the FORWARD/DOWNWARD action

Lab 3B-Task 2

1. Door_UP bit = 1 -> Motor_UP True
-> Motor_UPWARD True

*** & Stop bit=0 & UP_limit bit=0 & Motor_DOWNWARD bit=0

1. Door_DN bit =1 -> Motor_DN True
-> Motor_DOWNWARD True

*** & Stop bit=0 & DN_limit bit=0 & Motor_UPWARD bit=0

1. Stop bit=1 -> System OFF

Before the motor activation, Indication lights first!

1. UP_limit Bit = 0 & DN_limit Bit = 0 -> AJAR True
2. UP_limit Bit = 1 -> Door_OPEN True
3. DN_limit Bit = 1 -> Door_CLOSED True

Lab 3B-Task 1

- Same as what you did in task4 lab2, the only difference is that you need to add another rung

Tags and addresses

IN-0 means local.8.I.Data.0 **start** is tag name

OUT-0 means local.9.O.Data.0 **system** is tag name

- For input, use **IN-0** as **start**, **IN-1** as **stop**, **IN-2** as **LS1**, **IN-3** as **LS2**
- For output, use **OUT-0** as **system**, **OUT-1** as **CR1**, **OUT-2** as **CR2**
- 3 rungs, 4 inputs, 3 outputs
- **rung 0 is for system**
- **rung 1 is for CR1**
- **rung 2 is for CR2**

Lab 3B-Task 2

5 inputs: 3 Buttons: use IN-0 as **door up**, IN-1 as **door down**, IN-2 as **stop door**

2 Switches to represent sensors: IN-4 as **up limit**, IN-5 as **down limit**

7 outputs: use OUT-0 as **door ajar**, OUT-1 as **door open**, OUT-2 as **door closed**, OUT-3 as **motor up**, OUT-4 as **motor down**, OUT-5 as **motor upward**, OUT-6 as **motor downward**

- Push **door up** button **once** will make the motor **keep** moving **upward**, and **motor up** also turns on.
- Push **door down** button **once** will make the motor **keep** moving **downward**, and **motor down** also turns on.
- Any of “**door stop**”, “**up limit**”, “**motor down**” can stop the up action
- Any of “**door stop**”, “**down limit**”, “**motor up**” can stop the down action
- (3 examine off instructions for each of the up/down action)
- **5 rungs, first three rungs are for status lights (up/down/ajar), the other two rungs are for door up/down actions**

Lab 3B-Task 2

